

# A NOTE ON

## A toy model of two countries

### Content of the Handout

|   |                                      |      |
|---|--------------------------------------|------|
| 1 | Introduction.....                    | p. 1 |
| 2 | The model .....                      | p. 1 |
| 3 | Measurement equations and data ..... | p. 6 |
| 4 | Determinacy of the equilibrium ..... | p. 7 |

## 1. Introduction

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This handout delves into the intricacies of a Real Business Cycle model in a two-country context, focusing on the role of exporters. The model captures the dynamics of consumption goods exchange across international markets, highlighting how these markets achieve equilibrium through the mechanism of real exchange rate adjustments. Central to this equilibrium process is the uncovered parity condition, which governs the real exchange rate. This overview sets the stage for a deeper understanding of international trade and market responses within a real business cycle framework.

This model is a derived version of a open economy model of [Backus et al. \(1992\)](#) and [Kollmann \(1996\)](#). Estimated versions are given in [De Walque et al. \(2005\)](#) and [Adolfson et al. \(2008\)](#).

## 2. The model

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Let us suppose that there is two economies  $i = \{H, F\}$  of relative size  $n_i$  respectively.

### 2.1 Households

Suppose a continuum  $j \in [0, n_i]$  of welfare maximizing household enjoying current and future consumption and dislike working with respective curvatures  $\sigma_i^C, \sigma_i^H > 0$ . The household maximizes the following stream

of utilities discounted at a fixed rate  $\beta_i$ :

$$\max_{\{C_{it}(j), b_{it}(j), b_{it}^*(j), H_{it}(j)\}} \sum_{\tau=0}^{\infty} \beta_i^{\tau} E_t \left\{ \frac{(c_{it+\tau}(j) - h_i C_{t-1+\tau})^{1-\sigma_i^C}}{1 - \sigma_i^C} - \chi_i \frac{h_{it+\tau}(j)^{1+\sigma_i^H}}{1 + \sigma_i^H} \right\} \quad (1)$$

where  $C_{i,t}$  is consumption subject to external habits  $h \in [0, 1)$ ,  $h_{i,t}$  is hours worked and  $\chi_i > 0$  is a shift parameter allowing to pin down the amount of hours worked in steady state.

The budget constraint in real terms reads as:

$$w_t h_{it}(j) + \frac{r_{it-1}}{\pi_{it}^C} b_{i \rightarrow it-1} + e_{it} \frac{r_{kt-1}}{\pi_t^C} b_{i \rightarrow kt-1}(j) + \Pi_{it}(j) = c_{it}(j) + b_{it}(j) + e_{it} b_{i \rightarrow kt}(j) + e_{it} \frac{P_{it}}{P_{it}^C} f(b_{i \rightarrow kt}(j)) + T_{it}(j), \quad (2)$$

where  $b_{i \rightarrow it}$  denotes the acquisition of domestic riskless bonds and  $b_{i \rightarrow kt}$  foreign ones adjusted by the exchange rate  $e_{it}$ . Households pay an cost on foreign bonds, this is deemed necessary to remove an unit root component in consumption dynamics in an open economy environment. Here, the cost function reads as  $f(b_{i \rightarrow kt}(j)) = 0.5 \chi_B (b_{i \rightarrow kt}(j) - \bar{b}_{i \rightarrow k})^2$ . Households also pay a non distorsive tax  $T_{it}(j)$  and receive a wage payment  $w_t h_{it}(j)$  and profits  $\Pi_{it}(j)$  from shareholding firms.

The Lagrangian problem reads as:

$$\begin{aligned} \mathcal{L}_t^H(j) = & \sum_{\tau=0}^{\infty} \beta_i^{\tau} \left[ \frac{(c_{it+\tau}(j) - h_i C_{t-1+\tau})^{1-\sigma_i^C}}{1 - \sigma_i^C} - \chi \frac{h_{it+\tau}(j)^{1+\sigma_i^H}}{1 + \sigma_i^H} \right] \\ & + \beta^{\tau} \lambda_{it+\tau}^c(j) \left[ w_{it+\tau} h_{it+\tau}(j) + \frac{r_{it-1+\tau}}{\pi_{it+1+\tau}^C} b_{i \rightarrow it-1+\tau}(j) + e_{t+\tau} \frac{r_{kt-1+\tau}}{\pi_{it+\tau}^C} b_{i \rightarrow kt-1+\tau}(j) \right] \\ & - \beta^{\tau} \lambda_{it+\tau}^c(j) \left[ c_{it+\tau} + b_{it+\tau}(j) + e_{it} b_{i \rightarrow kt+\tau}(j) + \frac{P_{it}}{P_{it}^C} e_{it} g(b_{i \rightarrow kt+\tau}(j)) \right] \end{aligned}$$

where  $\lambda_{it}^c(j)$  is a lagrangian multiplier.

First order conditions are:

$$c_{it}(j) : (c_{it}(j) - h_i C_{t-1})^{-\sigma_i^C} - \lambda_{it}^c(j) = 0 \quad (3)$$

$$b_{i \rightarrow it}(j) : \beta_i E_t \{ \lambda_{it+1}(j) \} r_{it} / E_t \{ \pi_{it+1}^C \} - \lambda_{it}^c(j) = 0 \quad (4)$$

$$b_{i \rightarrow kt}(j) : \beta_i E_t \{ \lambda_{it+1}(j) e_{it+1} \} r_{kt} / E_t \{ \pi_{it+1}^C \} - e_{it} \lambda_{it}^c(j) = 0 \quad (5)$$

$$h_{it}(j) : -\chi_i h_{it}(j)^{\sigma_i^H} + \lambda_{it}^c(j) w_{it} = 0 \quad (6)$$

We define the real exchange rate as:

$$rer_{it} = e_{it} \frac{P_{kt}^C}{P_{it}^C} \text{ for } \forall k \neq j \in \{H, F\} \quad (7)$$

The households allocates their consumption on domestically produced goods and also imports foreign goods. The consumption basket of the household:

$$c_{it}(j) = \left( \varphi_i^{\frac{1}{\mu}} c_{i \rightarrow it}(j)^{\frac{\mu-1}{\mu}} + \left( \varepsilon_{it}^X (1 - \varphi_i) \right)^{\frac{1}{\mu}} c_{i \rightarrow kt}(j)^{\frac{\mu-1}{\mu}} \right)^{\frac{\mu}{\mu-1}} \quad (8)$$

where  $\varphi_i = (1 - n_k \alpha_i^C)$  refers to the relative weight of home and foreign goods, which is a function of the size of the foreign economy  $n_k$ , and  $\alpha_i^C$ , a measure of trade openness (see Corsetti (2006) and Goldberg

and Tille (2008)). In addition,  $\varepsilon_{it}^X$  is an AR(1) shock process of the demand for foreign goods and can be interpreted as an import shock. Parameter  $\mu \geq 0$  is the substitution degree between home goods and foreign goods, when  $\mu$  is high, households are more sensitive to price differentials between home and foreign goods, and will reduce their imports.

The price index of goods reads as:

$$P_{it}^C = \left[ \varphi_i (P_{it})^{1-\mu} + (1 - \varphi_i) (e_{it} P_{kt})^{1-\mu} \right]^{\frac{1}{1-\mu}} \quad (9)$$

Optimal demands for home  $c_{it}$  and foreign  $c_{it}^F$  varieties are given by:

$$c_{i \rightarrow it}(j) = \varphi_i \left( \frac{P_{it}}{P_{it}^C} \right)^{-\mu} c_{it}(j) \text{ and } c_{i \rightarrow kt}(j) = \varepsilon_{it}^X (1 - \varphi_i) \left( \frac{e_{it} P_{kt}}{P_{it}^C} \right)^{-\mu} c_{it}(j) \quad (10)$$

Here,  $c_{i \rightarrow kt}(j)$  are imports.

## 2.2 Retailers

Final firms competitively pack different types of good varieties  $j$  drawn on an unitary continuum to produce an homogeneous good sold to households. This firm is necessary when introducing monopolistic competition as they allows to create an elastic demand curve that intermediate firms exploits to generate positive profits. Production of final firms employs the following CES packing technology:

$$Y_{it} = \left( n_i^{-1/\epsilon_i} \int_0^{n_i} y_{it}(j)^{(\epsilon_i-1)/\epsilon_i} dj \right)^{\epsilon_i/(\epsilon_i-1)} \quad (11)$$

where  $\epsilon_i > 1$  is the perfect substitutability between different types of varieties  $j$ . The economy converges to a competitive equilibrium for  $\epsilon_i \rightarrow 1$ , while for any  $\epsilon_i > 1$ , goods are imperfect substitutes which allows firms to rise their selling price above their marginal cost.

Final firms are perfectly competitive and maximize profits,  $P_{it} Y_{it} - P_{it}(j) y_{it}(j)$ , under their technology constraint in [Equation 11](#) to obtain the optimal demand constraint for the  $i$ -th good:

$$y_{it}(j) = \frac{1}{n_i} \left( \frac{P_{it}(j)}{P_{it}} \right)^{-\epsilon} Y_{it}. \quad (12)$$

While the zero profit condition reads as:  $P_{it} = (\frac{1}{n_i} \int_0^{n_i} P_{it}(j)^{1-\epsilon_i} dj)^{1/(1-\epsilon_i)}$ .

## 2.3 Wholesale firms

Let us now suppose that there is a large number of identical firms  $j \in [0, n_i]$ . They hire workers and combine them with capital (both produced by them and supplied by households). Firms' technology is given by:

$$y_{it}(j) = A_i \varepsilon_{it}^A h_{it}(j)^\alpha \quad (13)$$

where  $\alpha \in [0, 1]$  is labor intensity and  $\varepsilon_{it}^A$  is a technology shock and  $A$  a scaling technology constant.

Following [Heutel \(2012\)](#), emissions are given by:

$$x_{it}(j) = \psi_i (1 - \mu_{it}(j)) y_{it}(j)^{1-\phi}$$

where  $x_{it}(j)$  is the flow of CO2 emissions, with  $\phi \in [0, 1]$  the elasticity of emissions to GDP,  $\psi_i \geq 0$  the carbon intensity and  $0 \leq \mu_{it}(j) \leq 1$  the fraction of emission avoided.

Profits of each firm  $i$  are given by:

$$\Pi_{it}(j) = mc_{it}(j) \varepsilon_{it}^Z h_{it}(j)^\alpha - w_{it} h_{it}(j) - \tau_t x_{it}(j) - g(\mu_{it}(j)) y_{it}(j) \quad (14)$$

and abatement expenditures  $g(\mu) = \theta_1 \mu^{\theta_2}$  with  $\theta_1 \in [0, 1]$  and  $\theta_2 > 1$ , while emissions are subject to carbon tax  $\tau_t$  and  $mc_{it}$  is the marginal cost of production.

The Lagrangian faced by entrepreneurs reads as:

$$\mathcal{L}_{it}(j) = mc_{it}(j) \varepsilon_{it}^Z h_{it}(j)^\alpha - w_{it} h_{it}(j) - \tau_t x_{it}(j) - g(\mu_{it}(j)) y_{it}(j) + \varrho_{it}(j) [y_{it}(j) - \varepsilon_{it}^Z h_{it}(j)^\alpha] \quad (15)$$

where  $\varrho_{it}(j)$  is the lagrangian multiplier over the supply curve.

Which allows us to determine both wages through the marginal product of labor of each type of households as well as the optimal abatement:

$$h_{it}(j) : w_{it} - \varrho_{it}(j) \frac{y_{it}(j)}{h_{it}(j)} = 0 \quad (16)$$

$$\mu_{it}(j) : g'(\mu_{it}(j)) - \tau_t \psi_i y_{it}(j)^{1-\phi} = 0 \quad (17)$$

$$y_{it}(j) : mc_{it}(j) - g(\mu_{it}(j)) - \tau_t (1 - \phi) \psi_i (1 - \mu_{it}(j)) y_{it}(j)^{-\phi} - \varrho_{it}(j) = 0 \quad (18)$$

## 2.4 Price Setting

Once the marginal cost of goods has been determined, we can now deduce their markup over the marginal to determine their selling price. However, firms face price adjustment costs paid in terms of home goods  $P_t$  using a mechasim à la [Rotemberg \(1982\)](#). These are quadractic adjustment costs are given by:

$$\Delta_{it}^P(j) = \frac{\xi_i}{2} \left( \frac{P_{it}(j)}{P_{it-1}(j)} - \bar{\pi}_i \right)^2 \quad (19)$$

where  $\xi_i \geq 0$  is the cost of adjusting price above or below the steady state inflation rate, denoted  $\bar{\pi}$ .

The profit ( $\Pi_{it}$ ) optimization becomes dynamic, intermediate firms set prices facing nominal rigidities by maximizing the discount sum of real profits:

$$\max_{\{P_{it}(j)\}} E_t \sum_{\tau=0}^{\infty} \beta_i^\tau \frac{\lambda_{it+\tau}}{\lambda_{it}} \left[ \frac{P_{it+\tau}(j)}{P_{it+\tau}} y_{it+\tau}(j) - \varepsilon_{it+\tau}^P mc_{it+\tau} y_{it+\tau}(j) - Y_{it+\tau} \Delta_{it+\tau}^P(j) \right] \quad (20)$$

where  $mc_{it}$  is the real marginal cost and  $\varepsilon_{it}^P$  is an AR(1) cost push shock. Replacing the demand function for each type of goods,  $y_{it}(j) = (P_{it}(j) / P_{it})^{-\epsilon_i} Y_{it}$ , the profits become:

$$\max_{\{P_{it}(j)\}} E_t \sum_{\tau=0}^{\infty} \beta_i^\tau \frac{\lambda_{it+\tau}}{\lambda_{it}} \left[ \frac{P_{it+\tau}(j)}{P_{it+\tau}} \left( \frac{P_{it+\tau}(j)}{P_{it+\tau}} \right)^{-\epsilon} - \varepsilon_{it+\tau}^P mc_{it+\tau} \left( \frac{P_{it+\tau}(j)}{P_{it+\tau}} \right)^{-\epsilon} - \Delta_{it+\tau}^P(j) \right] Y_{it+\tau} \quad (21)$$

The first order condition solving the optimization problem is given by:

$$\left[ (1 - \epsilon_i) \frac{Y_{it}}{P_{it}} - \epsilon_i \varepsilon_{it}^P \frac{mc_{it}}{P_{it}} - \frac{\xi_i}{P_{it}(j)} (\pi_{it}(j) - \bar{\pi}_i) \right] Y_{it} + \frac{\xi_i}{P_{it}(j)} E_t \left\{ \beta_i \frac{\lambda_{it+1}}{\lambda_{it}} (\pi_{it+1}(j) - \bar{\pi}_i) \pi_{it+1}(j) Y_{it+1} \right\} = 0$$

Assuming symmetry between firms in their price setting decisions (*i.e.*  $P_{it} = P_{it}(j)$ ), the new keynesian phillips curve emerges:

$$(1 - \epsilon_i) + \epsilon_i \varepsilon_{it}^P mc_{i,t} - \xi_i \pi_{it}(j) (\pi_{it}(j) - \bar{\pi}_i) + E_t \left\{ \beta_i \frac{\lambda_{it+1}}{\lambda_{it}} \frac{Y_{it+1}}{Y_{it}} \xi_i \pi_{it+1}(j) (\pi_{it+1}(j) - \bar{\pi}_i) \right\} = 0 \quad (22)$$

## 2.5 Authorities

We use a [Taylor \(1993\)](#) rule-type to model the setting of the nominal interest rate by the monetary authority:

$$r_{it} = r_{it-1}^{\rho_i} \left[ \bar{r}_i \left( \frac{\pi_{it}}{\bar{\pi}_i} \right)^{\phi_i^\pi} \left( \frac{Y_{it}}{\bar{Y}_i} \right)^{\phi_i^y} \right]^{1-\rho_i} \varepsilon_{it}^R \quad (23)$$

where  $\rho_i \in [0, 1)$  is the smoothing coefficient of the monetary policy rule,  $\phi_i^\pi$  the reaction to the inflation gap,  $\phi_i^y$  the reaction to the output gap and  $\varepsilon_{it}^R$  is an AR(1) monetary policy shock. Steady states for interest rates, inflation and output are respectively  $\bar{r}_i$ ,  $\bar{\pi}_i$  and  $\bar{Y}_i$ .

The CO2 tax policy is given by:

$$\tau_{it} = \tau_{i0} \varepsilon_{it}^\tau$$

where  $\tau_{i0}$  is the steady state CO2 tax (trillions \$ per Gt CO2),  $\varepsilon_{it}^\tau$  is a policy shock.

In addition, there is a government which consumes domestic goods. This public spending  $G_{it}$  is exogenous, as it follows an AR(1) shock process  $\varepsilon_{it}^G$ , which is expressed as a constant portion of output:

$$G_{it} = g_i^y \bar{Y}_i \varepsilon_{it}^G.$$

The balance sheet reads as:

$$G_{it} = T_{it} + \tau_{it} E_{it},$$

where  $T_{it} = \int_0^{n_i} T_{it}(j) dj$  and  $E_{it} = \int_0^{n_i} e_{it}(j) di$ .

## 2.6 Equilibrium conditions

First, the nominal exchange rate and real exchange rates in both sides of each countries are related by:

$$e_{Ht} = 1/e_{Ft} \text{ and } rer_{Ht} = 1/rer_{Ft}$$

we can thus substitute for the foreign country and express our model in terms of  $e_{Ht}$  and  $rer_{Ht}$ .

In addition, the size of the total mass is 1, imposing  $n_H + n_F = 1$ . One can simply substitute:

$$n = n_H \text{ and } 1 - n = n_F$$

Resource constraints is given by the total demand for home goods from home and foreign households (including also adjustment costs)

$$\int_0^n y_{Ht}(j) dj = \int_0^n [c_{H \rightarrow Ht}(j) + g(\mu_{it}(j)) y_{it}(j) + e_t f(b_{H \rightarrow Ft}) + g_{it}(j)] dj + \int_n^1 c_{F \rightarrow Ht}(j) dj \quad (24)$$

The aggregation of this equation yields:

$$Y_{Ht} = \varphi_H p_{Ht}^{-\mu} C_{Ht} + g(\mu_{Ht}) Y_{Ht} + G_{Ht} + \frac{(1-n)}{n} \varepsilon_{Ft}^X (1 - \varphi_F) \left( \frac{p_{Ft}}{rer_t} \right)^{-\mu} C_{Ft} + \frac{\chi_B}{2} \left( \frac{B_{H \rightarrow Ft}}{e_t} - \bar{B}_{H \rightarrow F} \right)^2$$

where  $B_{H \rightarrow Ft} = e_t b_{H \rightarrow Ft}$ ,  $p_{it} = \frac{P_{it}}{P_{it}^C}$ .

Similarly for the foreign country:

$$\int_n^1 Y_{Ft}(j) dj = \int_n^1 \left[ c_{F \rightarrow Ft}(j) + g(\mu_{Ft}(j)) y_{Ft}(j) + \frac{1}{e_t} f(b_{F \rightarrow Ht}) + g_{Ft}(j) \right] dj + \int_0^n c_{H \rightarrow Ft} dj$$

That yields:

$$Y_{Ft} = \varphi_F p_{Ft}^{-\mu} C_{Ft} + g(\mu_{Ft}) Y_{Ft} + G_{Ft} + \frac{n}{(1-n)} \varepsilon_{Ht}^X (1 - \varphi_H) (p_{Ht} r_{er_t})^{-\mu} C_{Ht} + \frac{\chi_B}{2} (e_t B_{F \rightarrow Ht} - \bar{B}_{F \rightarrow H})^2$$

To get the accumulation of net foreign asset, we aggregate the household constraint and assume net zero supply of domestic bonds,  $b_{H \rightarrow Ht} = b_{F \rightarrow Ft} = 0$ . One can simply rewrite the nominal budget constraint in a recursive form:

$$n e_t b_{H \rightarrow Ft} - n e_t R_{Ft-1} b_{H \rightarrow Ft-1} = n W_{it} H_{it} - n P_{it}^C C_{it} + n T_{it} + n \Pi_{it} - n P_{it} f(b_{H \rightarrow Ft-1})$$

Recall that with a zero profit condition over inputs markets, we have,  $m_{c_{it}} Y_{it} = \tau_{it} X_{it} + g(\mu_{Ht}) Y_{Ht} + W_{it} H_{it}$ . We find after rewriting and injecting  $Y_{it}$ :

$$\begin{aligned} e_t b_{H \rightarrow Ft} - e_t R_{Ft-1} b_{H \rightarrow Ft-1} &= Y_{Ht} - P_{Ht}^C C_{Ht} - P_{it} g(b_{H \rightarrow Ft}) \\ &= P_{Ht} \left[ \varphi_H \left( \frac{P_{Ht}}{P_{Ht}^C} \right)^{-\mu} C_{Ht} + \varepsilon_{Ft}^X (1 - \varphi_F) \left( \frac{P_{Ht}}{e_t P_{Ft}^C} \right)^{-\mu} C_{Ft} \frac{1-n}{n} + g(\mu_{Ht}) Y_{Ht} + G_{Ht} \right] \end{aligned}$$

Note also that we express quantity  $b_{H \rightarrow Ft}$  in domestic currency, that is interpreted in international macro as net foreign assets,  $NFA_{Ht} = e_t b_{H \rightarrow Ft}$  as well as real interest rates  $R_{Ft-1} = r_{Ft-1} / \pi_{Ft}^C$ .

$$NFA_{Ht} = R_{t-1}^* \frac{rer_t}{rer_{t-1}} NFA_{Ht-1} + p_{Ht} \left( \varphi_H p_{Ht}^{-\mu} C_{Ht} + \varepsilon_{Ft}^X (1 - \varphi_F) \left( \frac{p_{Ht}}{rer_t} \right)^{-\mu} C_{Ft} \frac{1-n}{n} \right) - C_{Ht} \quad (25)$$

In a world populated by two countries, international financial markets are clearing through

$$n NFA_{Ht} + (1-n) NFA_{Ft} = 0 \quad (26)$$

### 3. Measurement equations and data

Measurements can be set as follows (do not use all):

$$\begin{aligned} g_{Ht}^Y &= \Delta \log Y_{Ht} \\ g_{Ft}^Y &= \Delta \log Y_{Ft} \\ g_{Ht}^C &= \Delta \log C_{Ht} \\ g_{Ft}^C &= \Delta \log C_{Ft} \\ g_{Ht}^X &= \Delta \log (1-n) c_{F \rightarrow Ht} \\ g_{Ft}^X &= \Delta \log n c_{H \rightarrow Ft} \\ g_t^E &= \Delta \log e_t \\ g_t^{\mathcal{RER}} &= \Delta \log rer_t \\ g_{Ht}^P &= \pi_{Ht} - \pi \\ g_{Ft}^P &= \pi_{Ft} - \pi \\ \mathcal{R}_{Ht} &= r_{Ht} - \bar{r}_H \\ \mathcal{R}_{Ft} &= r_{Ft} - \bar{r}_F \end{aligned}$$

The model can be estimated on standard variables (output, consumption, inflation, nominal rate) and specific variable that is here the exports, real exchange rate or nominal exchange rate. Pay attention not to include too many observable variables. If you have 3 domestic variables, and eventually one foreign variable, you are already above the bar.

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## 4. Determinacy of the equilibrium

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For determinacy purpose, some variables need to be rewritten to be stationnary. In particular, prices and the exchange rate are not stationnary, thus requiring to express (i) the nominal exchange rate is variations  $\Delta e_t$ , and (ii) prices in a relative fashion  $p_{it} = P_{it}/P_{it}^C$ .

First, the relative price index can be expressed in growth rates to be stationnary:

$$\frac{p_{it}}{p_{it-1}} = \frac{\pi_{it}}{\pi_{it}^C} \quad (27)$$

Second, recall that  $rer_{it} = e_{it}P_{kt}^C/P_{it}^C$ , then we can express the real exchange in terms of growth too:

$$\frac{rer_{it}}{rer_{it-1}} = \Delta e_{it} \frac{\pi_{kt}^C}{\pi_{it}^C} \quad (28)$$

And the FOC on foreign bonds can also be written in terms of variations of the nominal exchange rate:

$$\mathbb{E}_t \{\Delta e_{it+1}\} = \left(1 + \chi_B rer_{it} (b_{H \rightarrow F} - \bar{b}_{H \rightarrow F})\right) \frac{R_{it}}{R_{kt}} \quad (29)$$

Third, dividing the CES price index by  $P_{it}^C$ , the following relation emerges:

$$1 = \left[\varphi_i p_{it}^{1-\mu} + (1 - \varphi_i) rer_{it}^{1-\mu C}\right]^{\frac{1}{1-\mu}} \quad (30)$$

Finally, to get a balanced uncovered interest rate parity (IUP), one needs to impose:

$$\bar{\pi} = \bar{\pi}_H = \bar{\pi}_F \text{ and } \beta = \beta_H = \beta_F$$

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